

1. Oersted Experiment & Magnetic Field

- * Hans Christian Oersted (1820) discovered magnetic effect of current.
- * Deflection in compass needle near current-carrying wire.
- * Magnetic field line direction around straight conductor.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Magnetic field is a vector quantity (has magnitude & direction).
- > Field lines emerge from North Pole & enter South Pole outside magnet.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

2. Right-Hand Thumb Rule & Field Patterns

- * Thumb points in direction of current, curled fingers give magnetic field direction.
- * Field lines around straight wire are concentric circles.
- * Field strength $B \sim I$, $B \sim 1/r$.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Closer field lines indicate stronger magnetic field.
- > Magnetic field lines never intersect each other.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

3. Magnetic Field due to Circular Loop & Solenoid

- * At center of loop, field lines appear as straight lines.
- * Solenoid: Long coil of many circular turns of insulated copper wire.
- * Field inside solenoid is uniform (parallel straight lines).

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Solenoid behaves like a bar magnet with N & S poles.
- > Strength of solenoid $B \sim$ number of turns (n) and current (I).

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

4. Electromagnet & Permanent Magnet

- * Electromagnet: Soft iron core placed inside solenoid.
- * Temporary magnetism: magnetic field disappears when current stops.
- * Permanent magnet: Steel core retains magnetism.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Electromagnet strength can be controlled easily.
- > Used in electric bells, cranes, magnetic levitation trains.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

5. Force on Current Conductor in Magnetic Field

- * Current-carrying conductor experiences force in magnetic field.
- * Force is maximum when direction of current is perpendicular to magnetic field.
- * $F = B I l \sin \theta$.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Fleming Left Hand Rule determines direction of force/motion.
- > Thumb: Force/Motion, Forefinger: Magnetic Field, Middle Finger: Current.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

6. Electric Motor & D.C. / A.C. Motor Principles

- * Electric Motor converts Electrical Energy to Mechanical Energy.
- * Commutator (Split Rings) reverses current direction every half rotation.
- * Armature coil rotates continuously in magnetic field.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Carbon brushes maintain sliding electrical contact.
- > Commercial motors use electromagnet & large number of coil turns.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

7. Electromagnetic Induction & Faraday Law

- * Production of electric current in coil due to changing magnetic field.
- * Discovered by Michael Faraday (1831).
- * Fleming Right Hand Rule gives direction of induced current.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Thumb: Motion, Forefinger: Magnetic Field, Middle Finger: Induced Current.
- > Used in Electric Generators (A.C. & D.C.).

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

8. Electric Generator & Domestic Circuits

- * Electric Generator converts Mechanical Energy into Electrical Energy.
- * A.C. Generator uses Slip Rings, D.C. Generator uses Split Rings.
- * Domestic wiring: Live wire (Red, 220V), Neutral wire (Black, 0V), Earth wire (Green).

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Live wire potential difference = 220V in India.
- > Earth wire provides low resistance path for safety against electric shock.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

9. Fuse Wire, Short Circuit & Overloading

- * Short Circuit: Direct contact between Live & Neutral wires causes huge current.
- * Overloading: Too many high-power appliances connected to single socket.
- * Fuse Wire / MCB protects electric appliances from excessive current.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Fuse wire connected in series with Live wire.
- > Fuse wire melts when current exceeds rated limit.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

10. One Page Quick Revision & Exam Booster

- * Summary of Oersted, Fleming Left/Right Rules, Right Hand Thumb Rule.
- * Comparison between A.C. & D.C. current.
- * Previous Years Board Questions & Diagram based questions.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > A.C. frequency in India = 50 Hz (changes direction every 1/100 sec).
- > Key safety measures in home electrical circuits.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.